

RISHI KIRAN KARNATAKAM

8328388715 • karnatakamrishi@gmail.com •
linkedin.com/in/rishi-kiran-karnatakam • github.com/Rishikarnatakam

Summary

Motivated Computer Science student seeking job and internship opportunities to apply my skills in software development and problem-solving. Quick to learn new technologies and adapt to challenges, focusing on contributing to innovative and impactful projects.

Education

NNRESGI **Dec 2021 – Present**
Bachelor of Technology in Computer Science and Engineering (GPA: 8.4)

KMIIT **Sep 2019 – Jun 2021**
Intermediate (10+2 equivalent) (GPA: 9.4)

SAGHS **May 2019**
Secondary School Certificate (GPA: 9.8)

Skills

- **Programming Languages:** C, Python, Dart, C#
- **Databases:** MySQL, MongoDB
- **Cloud Platforms:** Amazon Web Services (AWS)
- **Machine Learning/AI:** TensorFlow, TensorFlow Lite
- **Version Control:** Git, GitHub
- **Frameworks/Libraries:** Flask, Tkinter, Chess Library, Unity Engine, Flutter
- **DevOps Tools:** Jenkins

Projects

AI-Integrated Plant Disease Detection Mobile App **Apr 2025**

- Built a mobile app for real-time plant disease detection using camera or gallery images.
- Integrated a TensorFlow Lite model (Inception V3) for accurate disease prediction with offline support.
- Enabled local data storage and automatic syncing to MongoDB Atlas when connected online.
- Designed a clean, modern UI with a sidebar to view history, profile, and AI predictions.

- **Technologies:** Flutter, Dart, TensorFlow Lite, MongoDB Atlas

AI-Powered Chess Engine with Genetic Algorithm Optimization

Aug 2024

- Developed an AI-driven chess engine utilizing minimax, alpha-beta pruning, and Genetic Algorithms for optimized decision-making.
- Enhanced strategic depth and move accuracy by 20%.
- Achieved a 30% improvement in move generation speed through alpha-beta pruning.
- **Technologies:** Python, Tkinter, Chess Library

Cotton Plant Disease Detection and Classification Using Transfer Learning

Feb 2024

- Developed a machine learning model to classify various diseases on cotton leaves through image processing techniques.
- Improved disease detection accuracy by 25% using transfer learning with pre-trained CNN models.
- Achieved 90%+ accuracy on test datasets, optimizing performance for agricultural disease diagnosis.
- **Technologies:** Python, TensorFlow

Interactive Solar System Model and 3D Game Development

Oct 2023

- Developed a 3D interactive solar system model to enhance gamified learning.
- Created a first-person 3D game inspired by Flappy Bird, leveraging dynamic renderings and immersive gameplay elements.
- **Technologies:** C#, Unity Engine

Awards

- **National Hackathon on Generative AI** – Achieved 2nd place in a National Level Hackathon focused on Generative AI. (Sep 2024)
- **Internal Hackathon on Generative AI** – Achieved 3rd place in a college-hosted Hackathon focused on Generative AI. (Aug 2024)
- **National Hackathon on AR/VR Development** – Ranked in the top 10 at a national-level hackathon focused on AR/VR. (Oct 2023)

Certifications

AWS Academy Graduate - AWS Academy Cloud Architecting
AWS Academy

Sep 2024

AICTE Virtual Internship

2024

Nalla Narasimha Reddy Education Society's Group of Institutions

Python For Data Science

IIT Madras (NPTEL)

Jan 2025

The Joy of Computing Using Python

IIT Madras (NPTEL)

Jul 2023

Supervised Machine Learning: Regression and Classification

Stanford University (Coursera)

Jun 2023